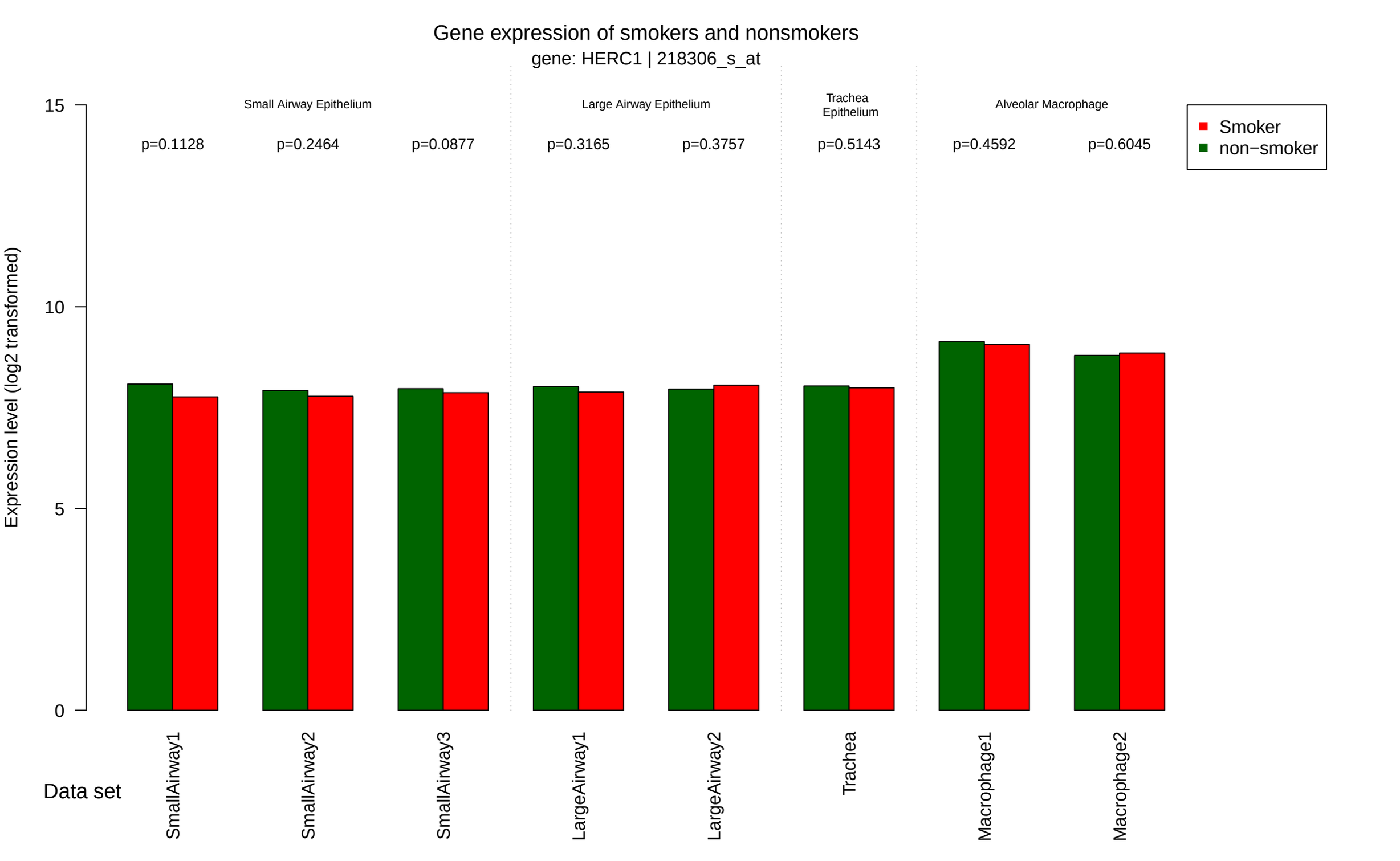
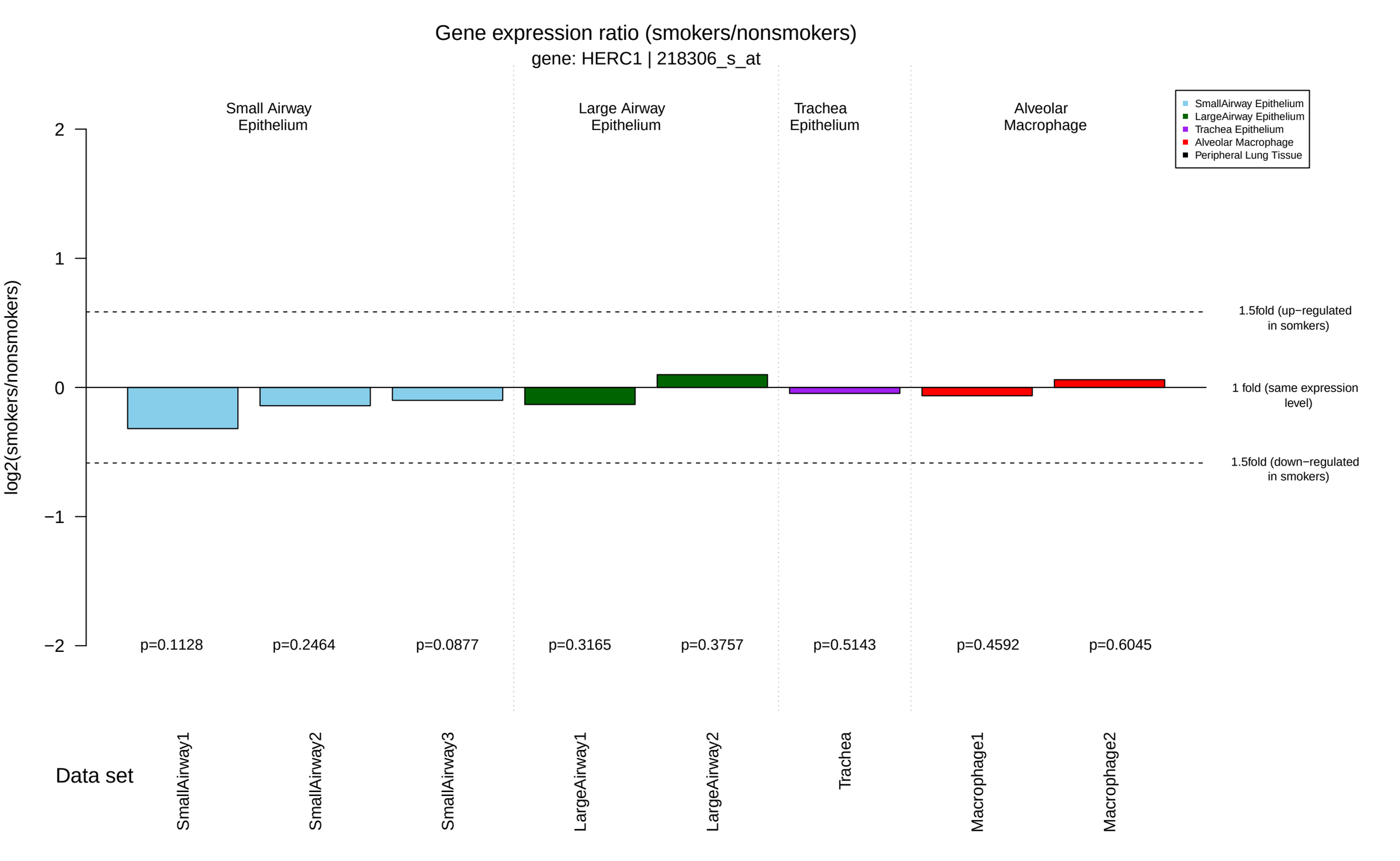
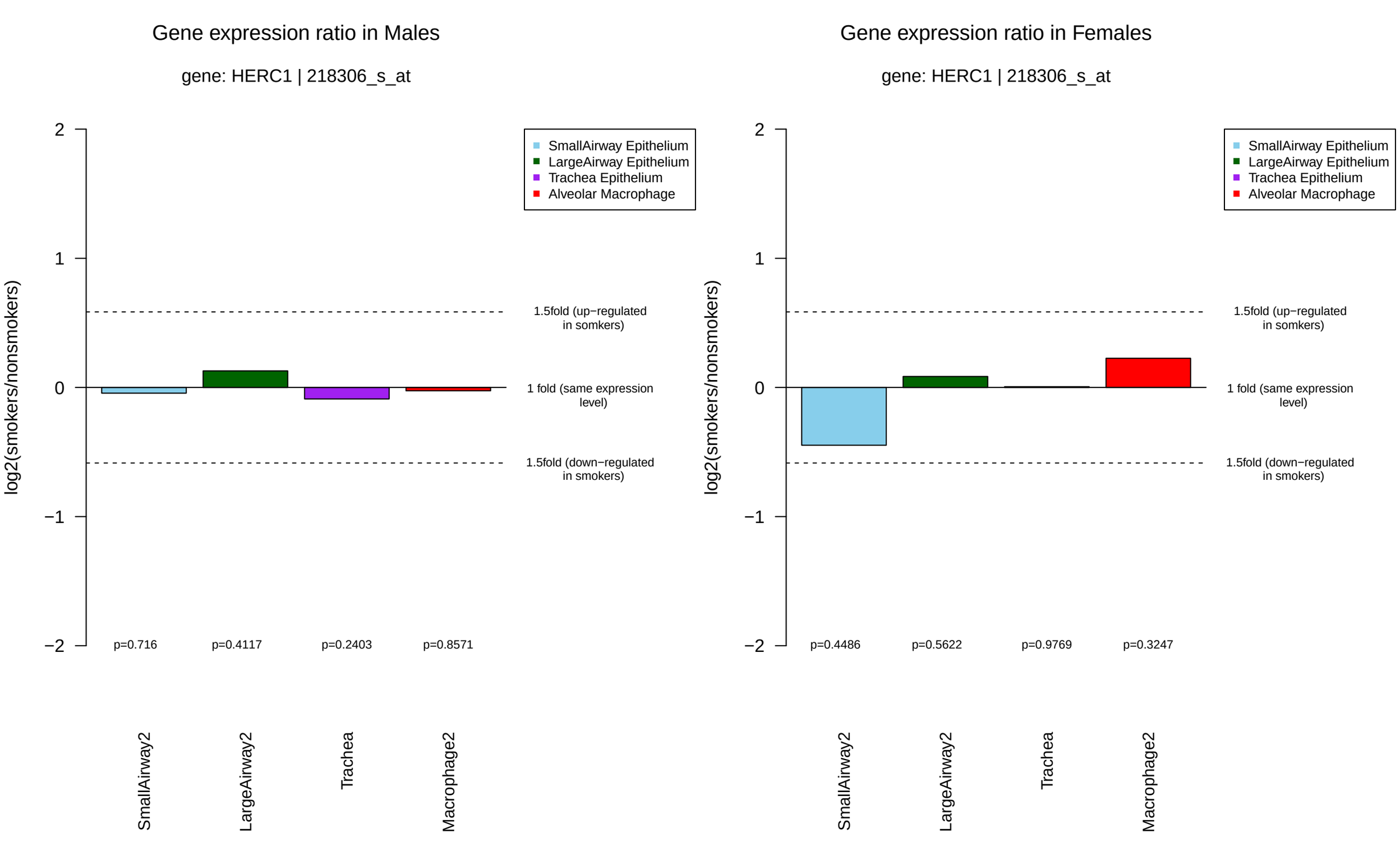
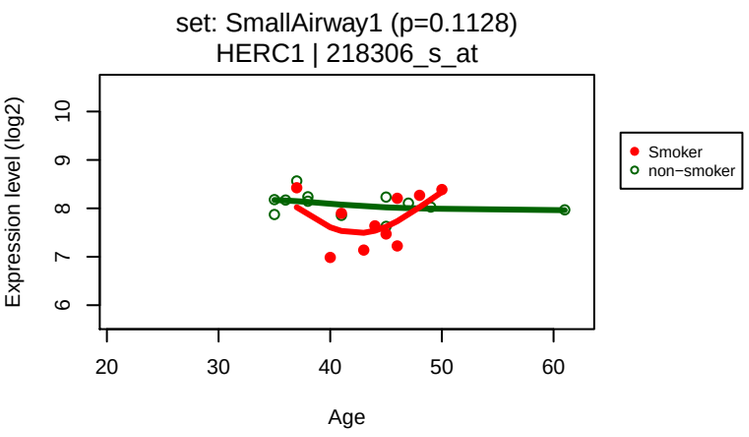
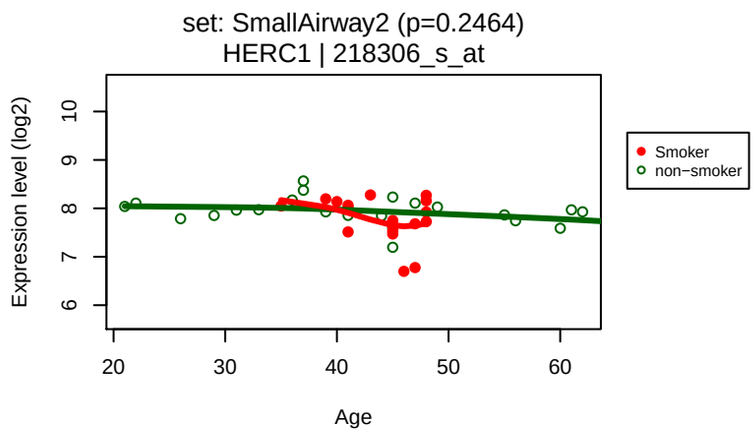
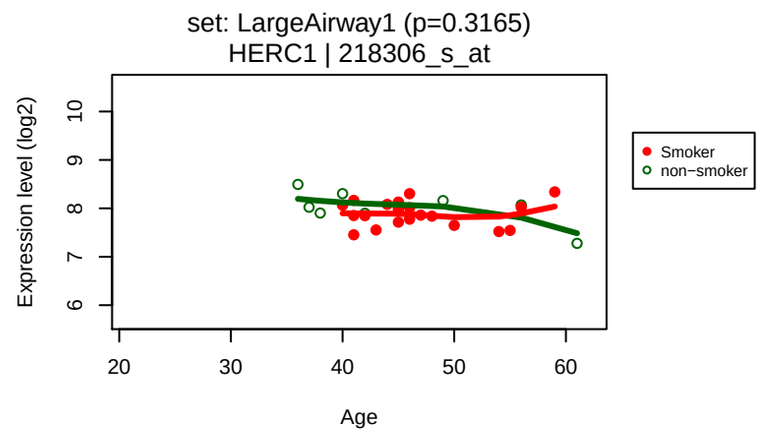

Gene expression of smokers and nonsmokers
gene: HERC1 | 218306_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1128
p=0.2464
p=0.0877
p=0.3165
p=0.3757
p=0.5143
p=0.4592
p=0.6045
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

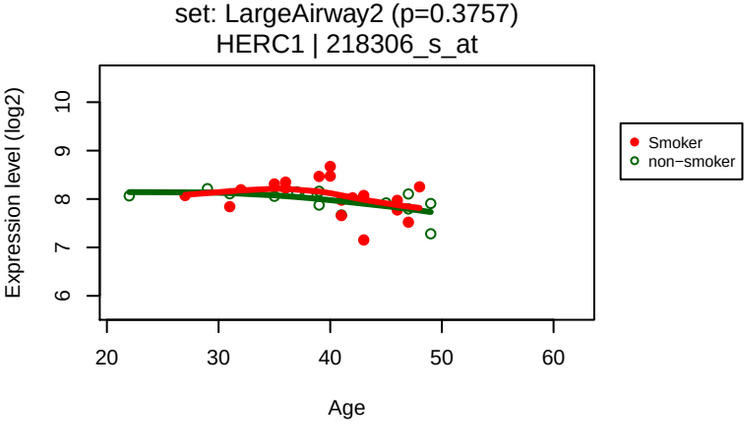

Gene expression ratio (smokers/nonsmokers)
gene: HERC1 | 218306_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1128
p=0.2464
p=0.0877
p=0.3165
p=0.3757
p=0.5143
p=0.4592
p=0.6045
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

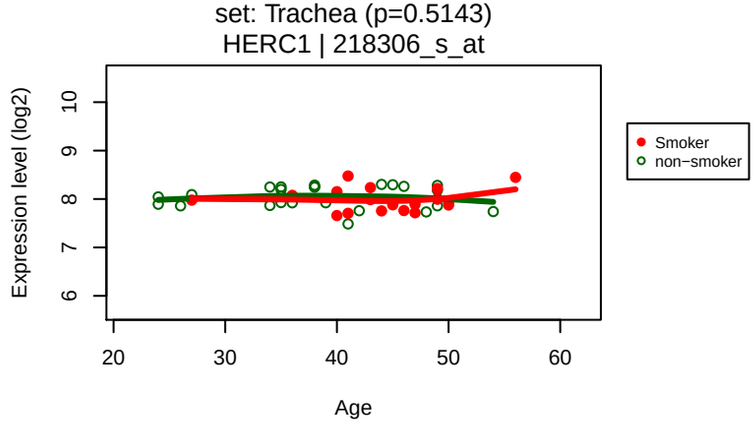

Gene expression ratio in Males
gene: HERC1 | 218306_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.716
p=0.4117
p=0.2403
p=0.8571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HERC1 | 218306_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4486
p=0.5622
p=0.9769
p=0.3247
SmallAirway2
LargeAirway2
Trachea
Macrophage2

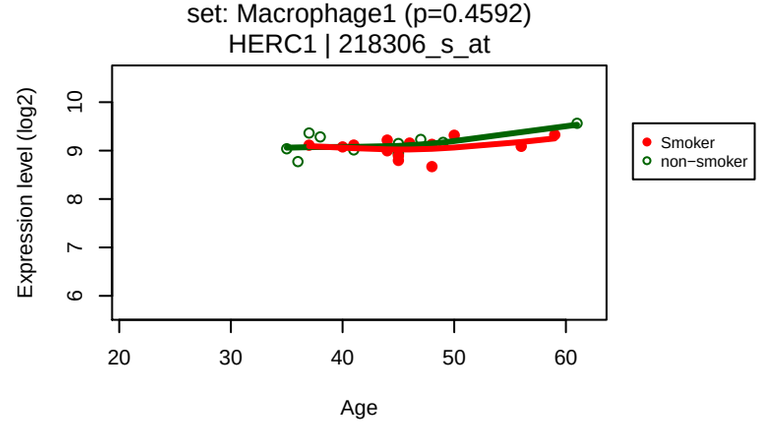

set: SmallAirway1 (p=0.1128)
HERC1 | 218306_s_at
Expression level (log2)
Age
Smoker
non−smoker

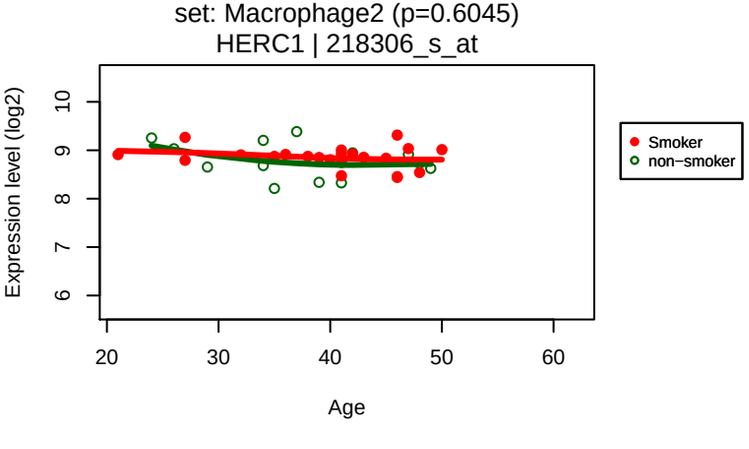

set: SmallAirway2 (p=0.2464)
HERC1 | 218306_s_at
Expression level (log2)
Age
Smoker
non−smoker

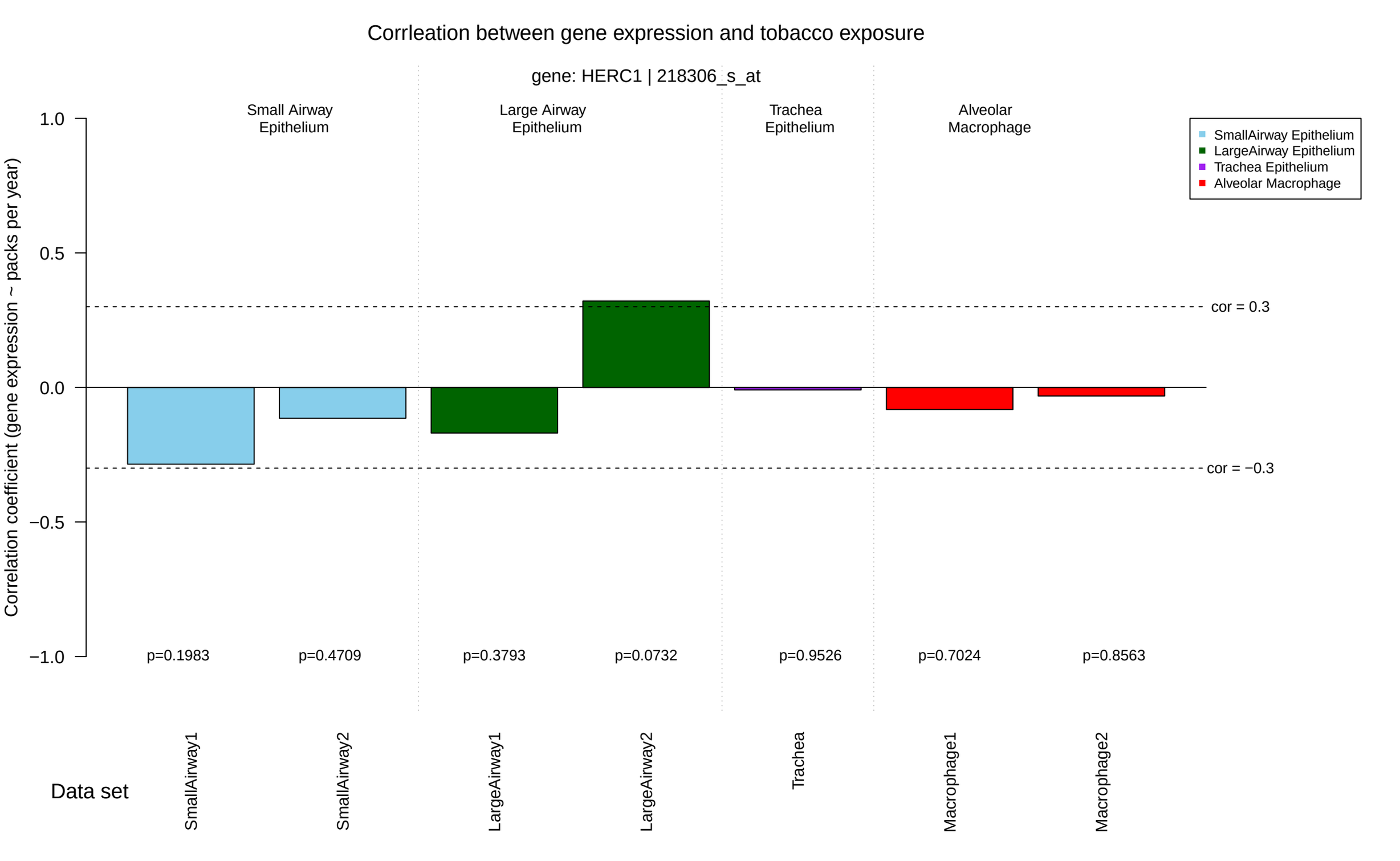

set: LargeAirway1 (p=0.3165)
HERC1 | 218306_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3757)
HERC1 | 218306_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5143)
HERC1 | 218306_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4592)
HERC1 | 218306_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6045)
HERC1 | 218306_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HERC1 | 218306_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1983
p=0.4709
p=0.3793
p=0.0732
p=0.9526
p=0.7024
p=0.8563
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2